SwiftLog Logistics Limited

Supply Chain Intelligence & Feature Engineering for Predictive Logistics

Technical Report · Mission DS05 · Data Science Track · Week 5

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Data: 12 SwiftLog tables, 5,000 orders, 1 Jan – 8 Sep 2026

1. Executive Summary

SwiftLog asked what its operational data says about supply chain performance and what would be needed for predictive logistics. We cleaned and linked 12 tables (about 36,115 rows), built one order-level analytical table, ran 29 visual analyses, 10 hypothesis tests, 80+ engineered fields, three segmentations and a baseline machine-learning test.

Headline findings

- Only 33.2% of orders are Delivered; 32.5% are Cancelled (7.23M of order value) and 34.3% are Pending. Confirmed (non-cancelled) order value is 14.44M against 21.67M gross.
- 63.7% of shipments are delayed by 3 or more days (mean 3.57 days). Delay is spread evenly from 0 to 7 days and is statistically the same across warehouses (p = 0.33), regions (p = 0.10), weekdays and order types. This is a network-wide problem, not a problem-warehouse issue.
- Order status contradicts the shipment and return records: 1,626 shipments sit on Cancelled orders. Delivery KPIs cannot be fully trusted until this is reconciled.
- Revenue is flat (about 1.74M a month) and evenly spread: the top 20% of customers deliver 41% of revenue and no product group dominates.
- Supplier performance cannot be analysed: no table links suppliers to products, stock or shipments. We report this as a data gap rather than inventing a link.
- None of the 10 hypothesis tests was significant at 0.05, and the best predictive model reached AUC 0.52 (0.50 is chance). Today's fields cannot predict delays or returns; new data must be captured first.

Top recommendations: reconcile order status with shipment events; run a network-wide on-time delivery programme; attack the cancellation and pending leak; rebalance warehouse staffing; run a win-back programme for lapsing high-value customers; and capture supplier, carrier and date fields before building ML.

2. Data Understanding

The data set contains 12 CSV tables (Customers, Products, Suppliers, Warehouses, Employees, Orders, Order_Items, Inventory, Shipments, Delivery_Performance, Returns, plus supplied Orders). Orders is the hub: customers, line items, shipments and returns hang off OrderID, and warehouses link shipments, stock and staff. The Suppliers table is isolated.

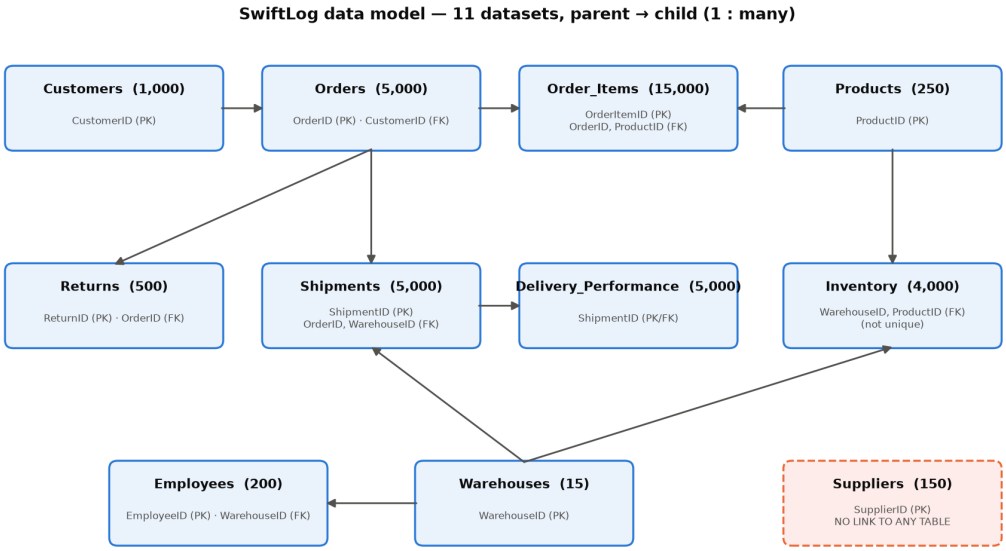


Figure 1. Data relationship diagram. Orders is the hub: customers, line items, shipments and returns all hang off OrderID. Warehouses connect shipments, stock and staff. Suppliers is isolated.

***Business implication:** Operational questions (delivery, returns, customer value) can be joined end-to-end, but supplier performance cannot — SwiftLog must add a SupplierID to Products or Shipments to evaluate suppliers.*

Key integrity checks: no missing values in any table; no orphan foreign keys (0 found); Delivery_Performance agrees with Shipments on all delay values; and the delivery status is a pure rule of DelayDays (0–2 On Time, 3+ Delayed). The checks below show what the data does not contain.

Check	Result
Missing values (all tables)	0
Exact duplicate rows — Inventory	3
Inventory rows sharing a (Warehouse, Product) key with another row	2602
Order_Items lines repeating an (Order, Product) pair	96
Orders with no line items	236
Shipments recorded for CANCELLED orders	1626
Shipments recorded for PENDING orders	1716
Returns recorded against non-Delivered orders	335
Orders with more than one return record	27
Shipments.DelayDays ≠ Delivery_Performance.DelayDays	0
Delivery status is a pure rule of DelayDays (≤2 On Time, ≥3 Delayed)	0
Orders placed in the partial month (Sep 2026, data ends 8 Sep)	154
Supplier key present on any operational table	0
Warehouse capacity field	0
Ship / delivery dates (only OrderDate exists)	0
Data dictionary column-level definitions	0

A full automated profile of every table is in Reports/Data_Profiling_Report.html. ydata-profiling could not be installed in the build environment (no package index access), so an equivalent profile was generated with custom code: per-column type, missingness, distinct values, summary statistics and distributions.

3. Data Cleaning Process

All cleaning lives in reusable functions (standardise_text, parse_dates, iqr_flag, clean_all, build_order_master) so that the same pipeline can be re-run on a new extract. Every step is logged with rows affected and a reason. No rows were deleted other than exact duplicates.

Step	Action	Rows	Reason
1	Trim whitespace, unify case of text/ID fields (all tables)	0 changed	Prevents duplicate categories; data was already clean
2	Convert OrderDate to datetime; derive Year, Month, Quarter, DayOfWeek, YearMonth	5,000	Enables time analysis. Last order is 8 Sep 2026, so September is partial
3	Merge repeated (Order, Product) lines	96	Same product twice in an order is one basket line
4	Create LineValue = Quantity × UnitPrice	14,904	Revenue is not stored in the raw data
5	Drop exact duplicate Inventory rows	3	Identical repeated records
6	Sum stock across repeated (Warehouse, Product) keys	1,542	Assumed to be stock lots; sensitivity check gave rank correlation 0.82 so conclusions are largely stable
7	IsDelayed = DelayDays ≥ 3; DelaySeverity	5,000	Rule verified against supplied delivery status
8	Build order-level master table (5,000 × 31)	5,000	Single analytical grain
9	Flag IQR outliers; cap order value at the 99th percentile in a separate column	344 flags	Extreme but valid values kept; capped copy for sensitive statistics

Outliers were flagged, not removed: Order_Items.Quantity 23, LineValue 129, order value 106, consolidated stock 86. They are genuine large orders and stock positions. Categorical encoding is covered in Section 6. Twelve cleaned CSVs are saved in Data/Cleaned_Datasets.

4. Exploratory Data Analysis

Revenue definitions: gross = all orders (21.67M); confirmed = non-cancelled (14.44M), used as the default; realised = Delivered only (7.11M). The 23 charts below each carry an interpretation and business implication; all 29 charts are in Visualizations/Charts.

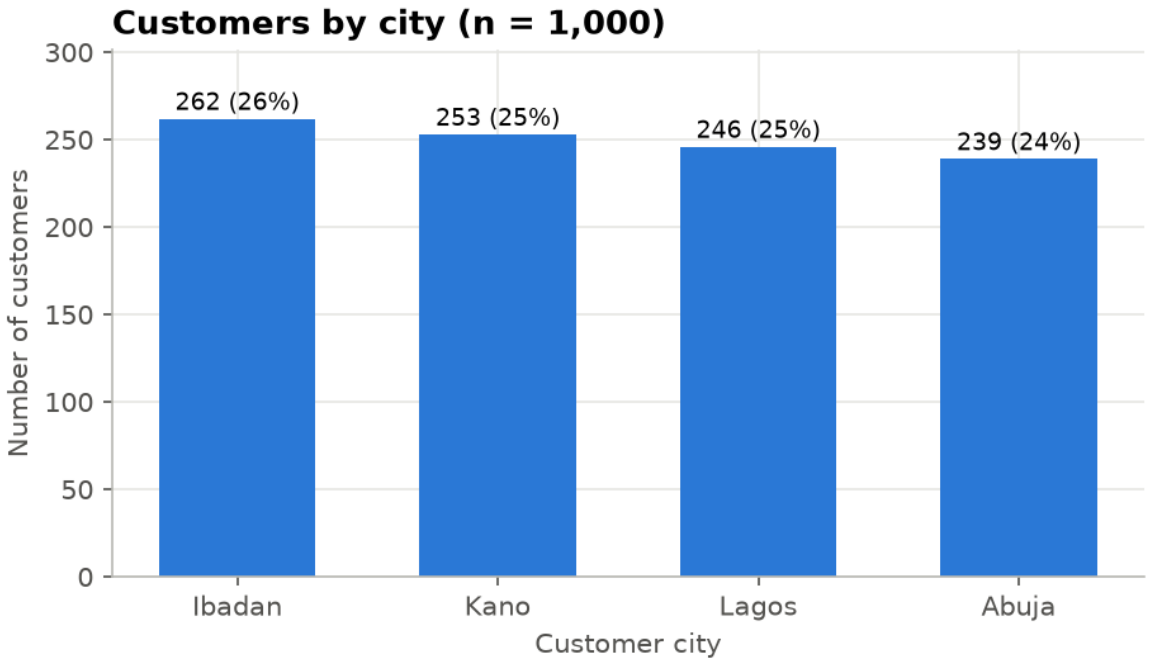


Figure 2. Customers by city. Customers split almost evenly across four cities (Ibadan largest at 26.2%, Abuja smallest at 23.9%).

Business implication: *Demand is geographically balanced, so no single city dominates planning — but Ibadan holds the largest customer base yet has no SwiftLog warehouse (all Ibadan orders are dispatched from another city).*

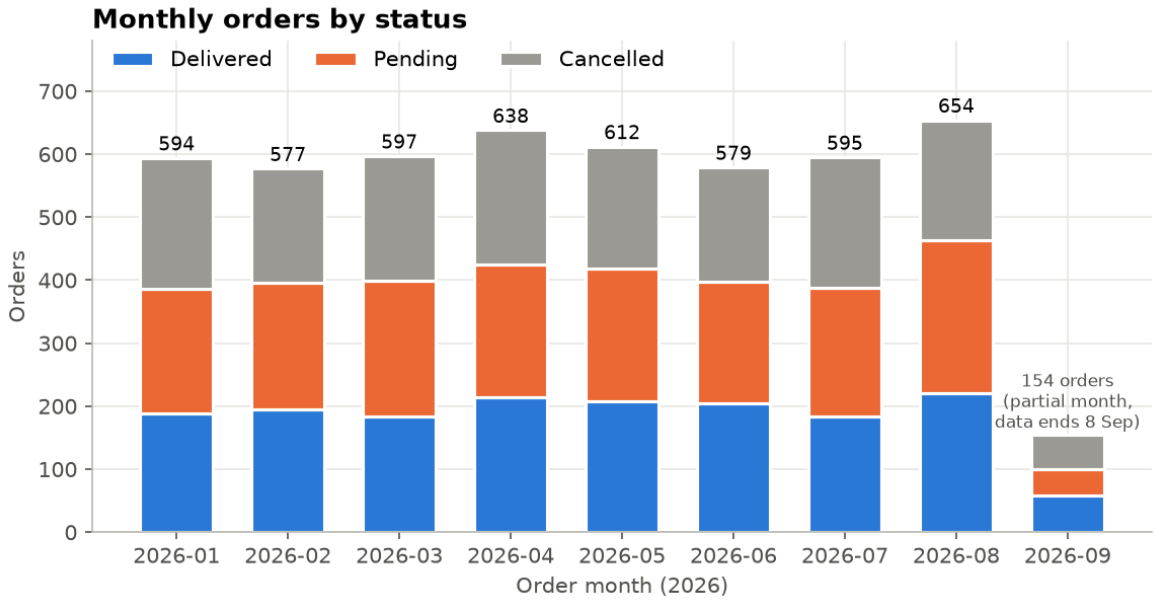


Figure 3. Monthly orders by status. Order volume is steady at roughly 606 orders a month (range 577–654; peak 2026-08, low 2026-02). In every month about a third of orders are Cancelled and a third are still Pending.

Business implication: *Only 33.2% of orders are Delivered while 32.5% are Cancelled — a persistent conversion leak worth 7.23M in order value. Fixing cancellations is a bigger lever than adding demand.*

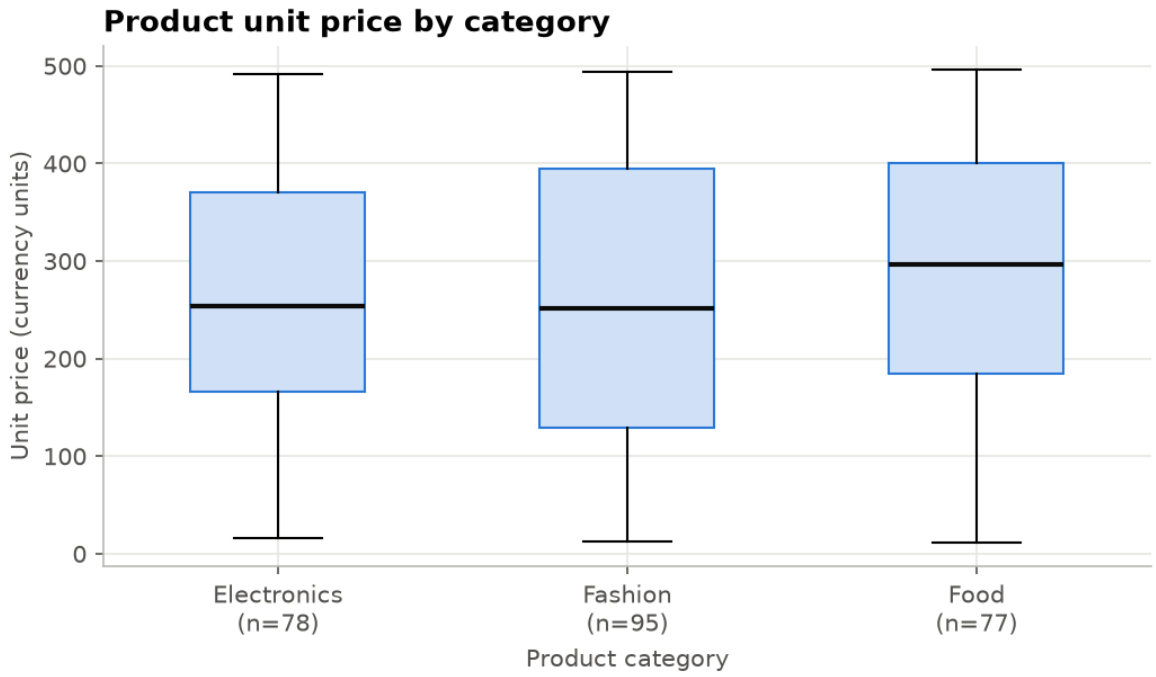


Figure 4. Unit price by category. All three categories span the full price range (≈12–497) with similar medians (251–297); category is not a proxy for price tier.

Business implication: *Pricing is not differentiated by category, so category-level revenue differences come from volume and mix, not premium pricing — a pricing-strategy opportunity (premium vs value tiers).*

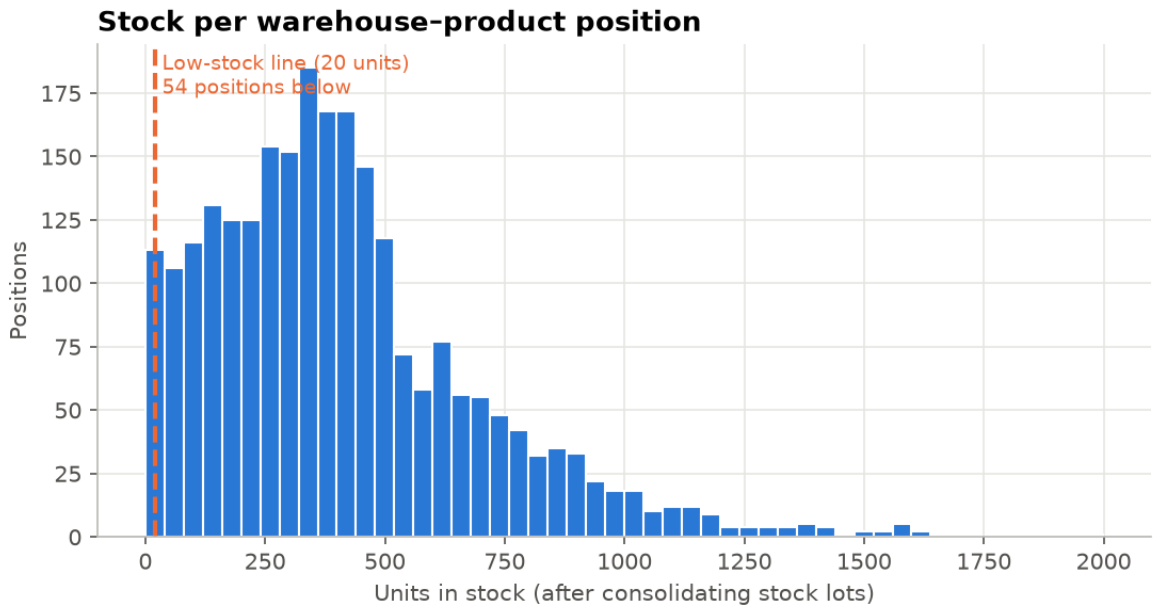


Figure 5. Stock distribution. Stock per position is spread broadly (median 364 units); only 54 of 2,455 positions (2.2%) fall below 20 units, 3 are at zero.

Business implication: *Stock-outs are rare overall but not absent: the few low/zero positions are concentrated and can be fixed with targeted replenishment rather than a blanket stock increase. The long right tail comes from consolidated stock lots.*

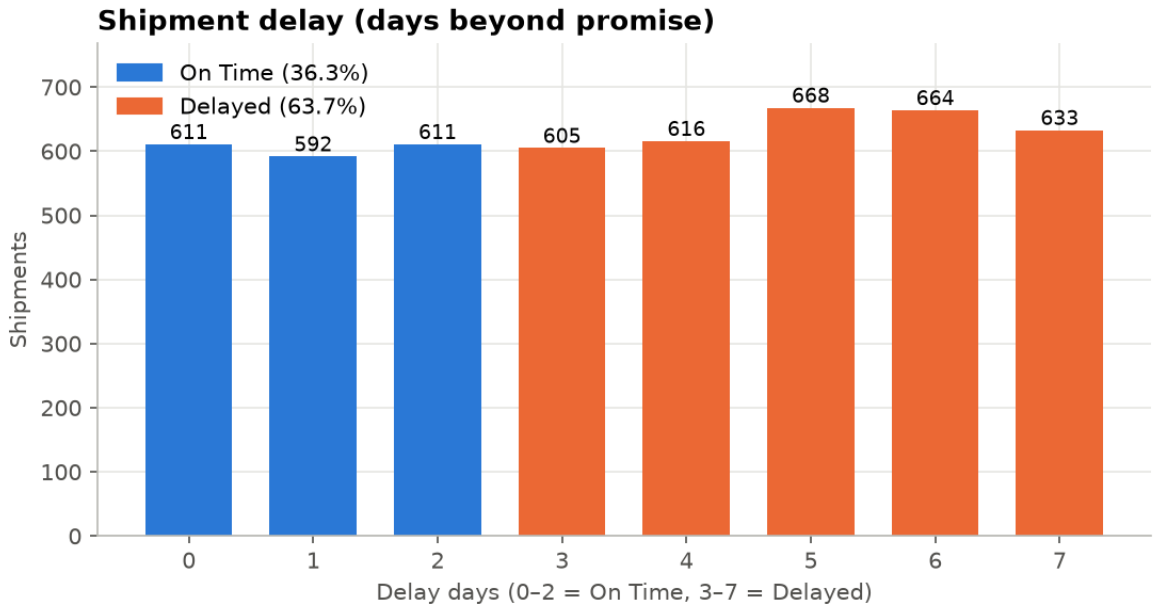


Figure 6. Distribution of shipment delay. Delays are spread almost uniformly from 0 to 7 days (no peak at zero). 63.7% of shipments are 'Delayed' (≥3 days); the mean delay is 3.57 days.

***Business implication:** A uniform spread means lateness is systemic, not the result of a few bad events: fewer than 4 in 10 shipments are on time. Delivery reliability needs a network-wide process fix, not firefighting.*

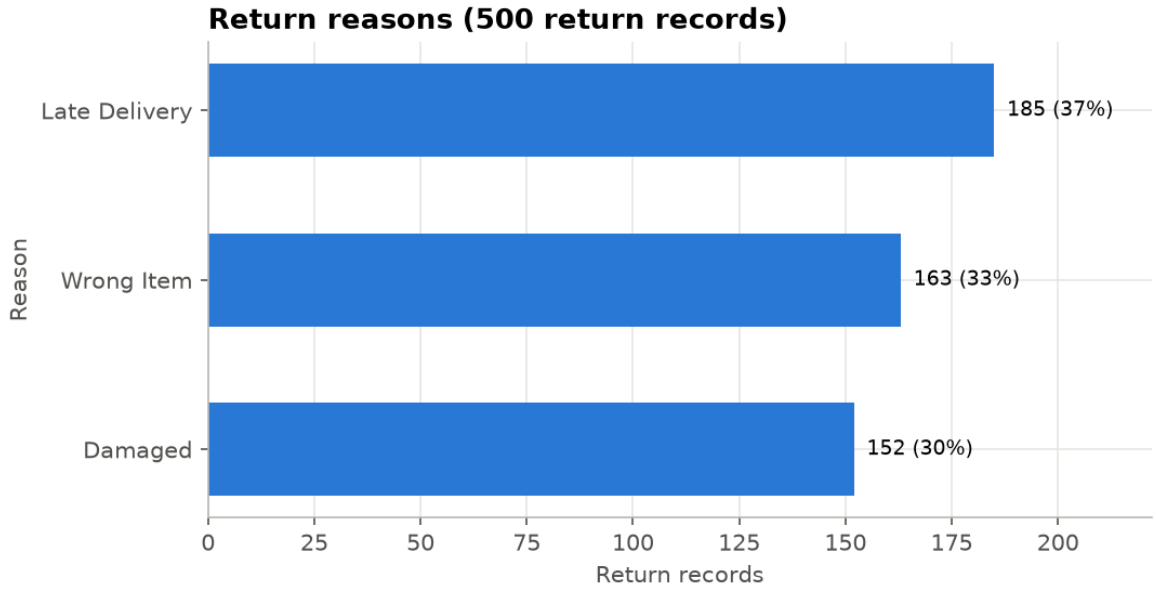


Figure 7. Return reasons. Late Delivery is the most frequent reason (37.0%), followed by Wrong Item (32.6%) and Damaged (30.4%); 9.5% of orders have at least one return.

***Business implication:** Roughly two-thirds of returns are controllable by operations (wrong item = picking accuracy; damaged = handling/packaging) rather than customer preference, so returns can be cut through warehouse process quality.*

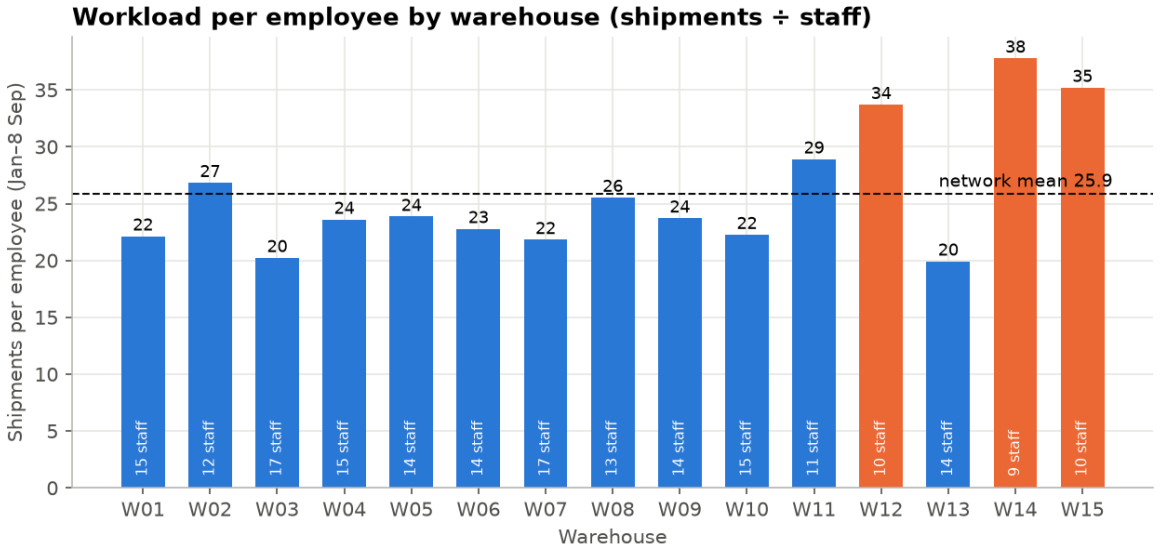


Figure 8. Warehouse workload per employee. Shipments handled per employee range from 20 (W13) to 38 (W14) — a 1.9× spread. Heaviest loads: W14, W15, W12.

Business implication: *Staffing is not aligned with workload: shipment volume is nearly equal across warehouses (≈280–370) but headcount varies 9–17. Rebalancing staff toward the overloaded sites is a low-cost capacity gain. (Warehouse capacity is not in the data, so this is a load proxy, not utilisation.)*

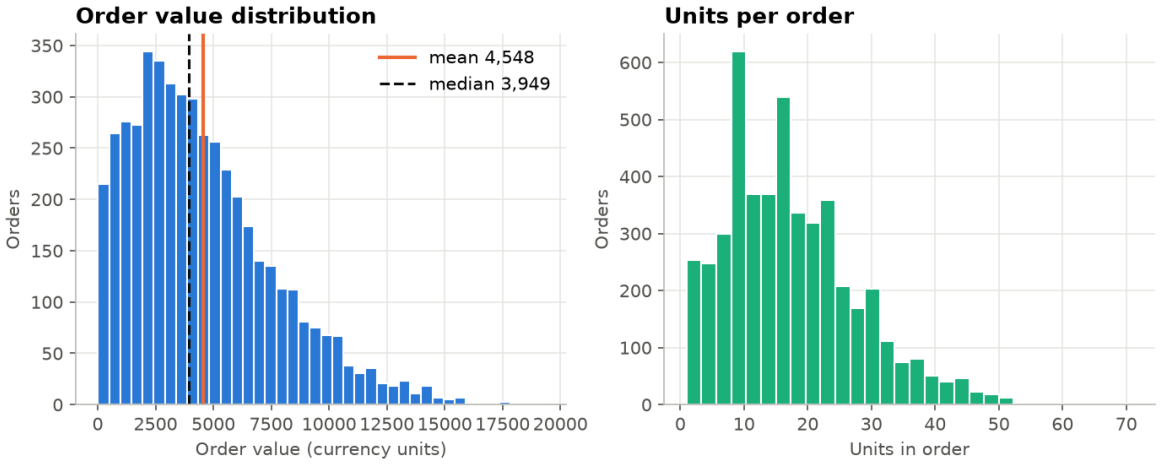


Figure 9. Order value and basket size. Order value is right-skewed (mean 4,548 > median 3,949; skew 0.99); a typical basket is 16 units across 3 products.

Business implication: *Baskets are large for a logistics catalogue, and 236 orders (4.7%) carry no line items at all — an order-capture defect that hides revenue and should be closed at source.*

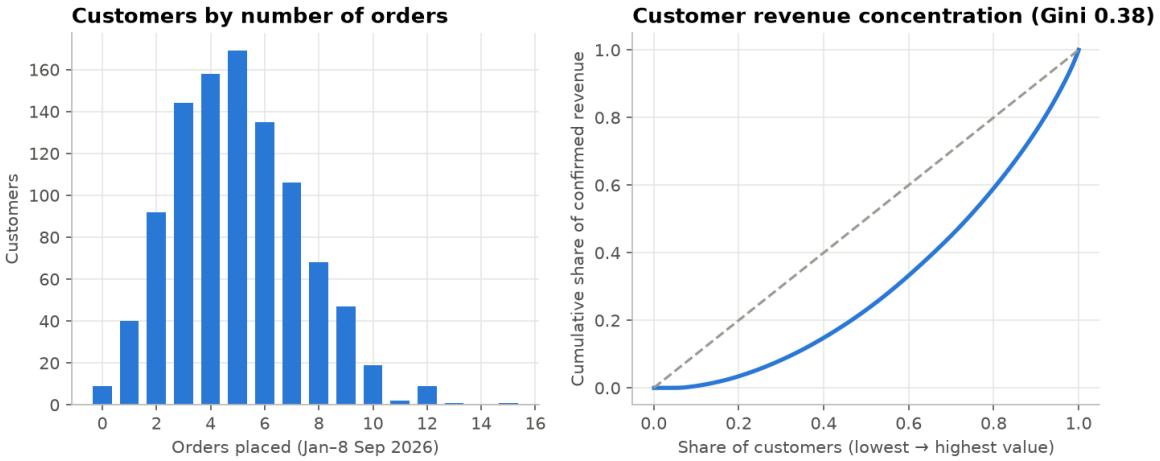


Figure 10. Customer order frequency and revenue concentration. Customers place 5.0 orders on average (max 15); 9 customers never ordered. Revenue is only mildly concentrated: the top 20% of customers deliver 41.0% of confirmed revenue (Gini 0.38; the Pareto 80/20 rule does not hold).

Business implication: *There is no small group of "whales" to protect — value comes from a broad, evenly-spread base. Retention and frequency programmes across the whole base will matter more than key-account management.*

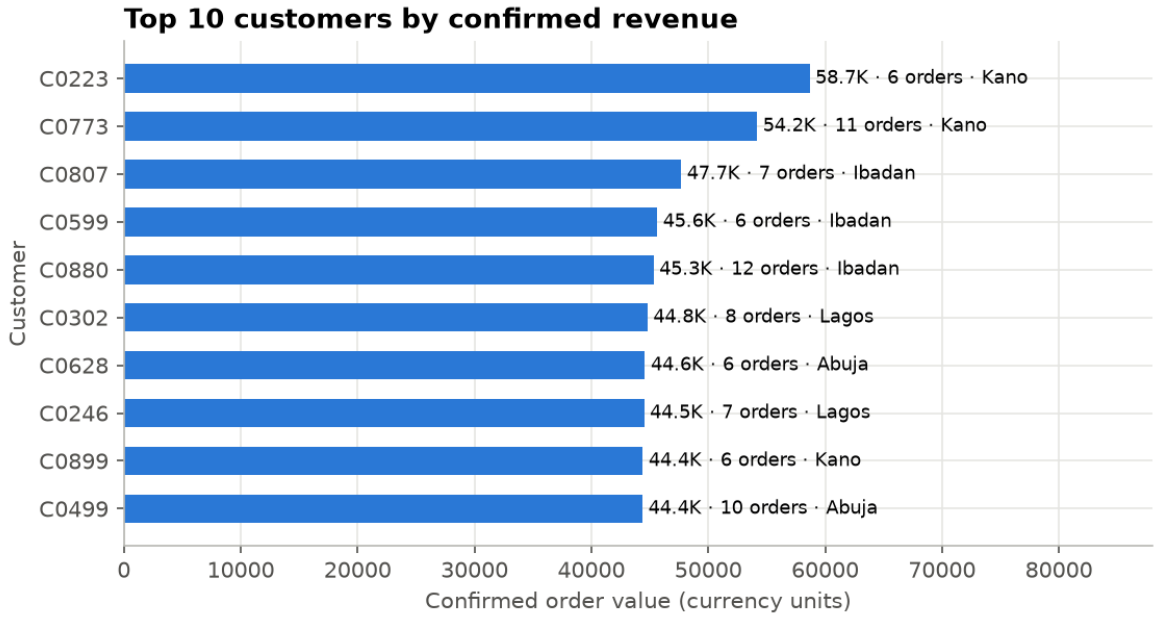


Figure 11. Top 10 customers. The top customer (C0223) generated 58.7K; together the top 10 account for only 3.3% of confirmed revenue.

Business implication: *Even SwiftLog's best customers are individually small, which limits single-customer risk but also means growth must come from the base, not a few accounts. Use these 10 as a pilot group for loyalty and service-level offers.*

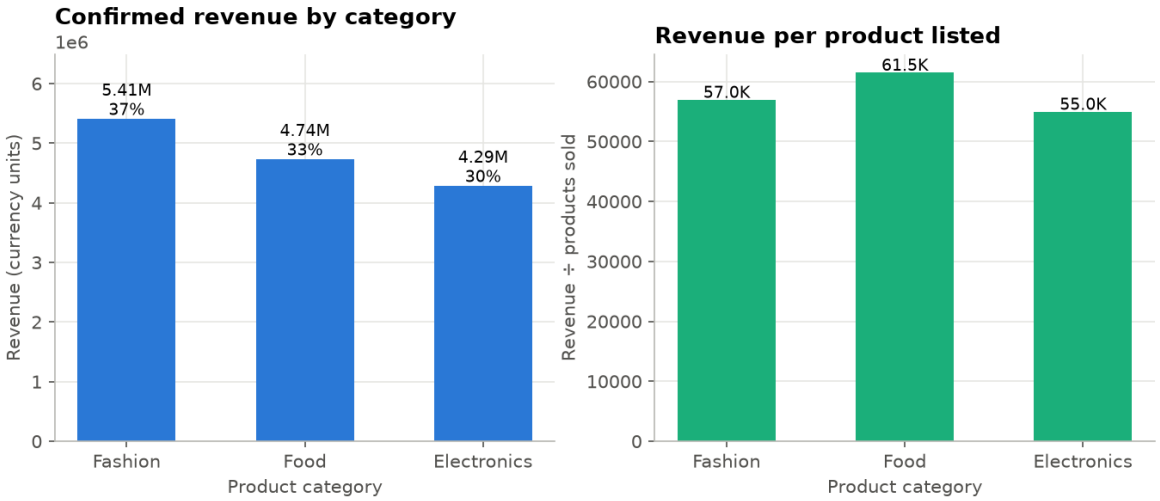


Figure 12. Category vs revenue. Fashion leads with 37.5% of confirmed revenue (5.41M), then Food (32.8%) and Electronics (29.7%). On a per-product basis Food is strongest.

***Business implication:** Fashion leads partly because it has the most SKUs (95 vs 77–78); on a per-SKU basis Food is the most productive range. Range-extension decisions should use revenue per SKU, not total revenue.*

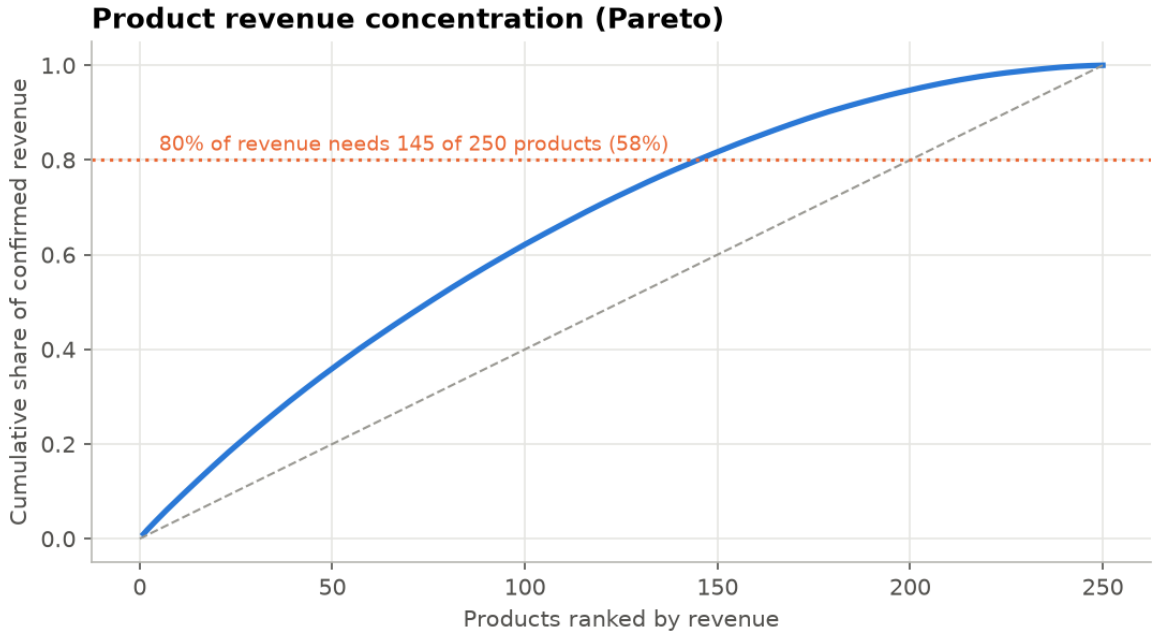


Figure 13. Product Pareto curve. Revenue is spread thinly across the catalogue: the top 20% of products generate only 36.0% of revenue, and 145 of 250 products (58%) are needed to reach 80%.

***Business implication:** The catalogue has no "hero" products and no obvious tail to cut. Best-seller lists (e.g. P085, P040, P025) are only marginally ahead, so range-rationalisation should be based on margin and return data, not revenue rank alone.*

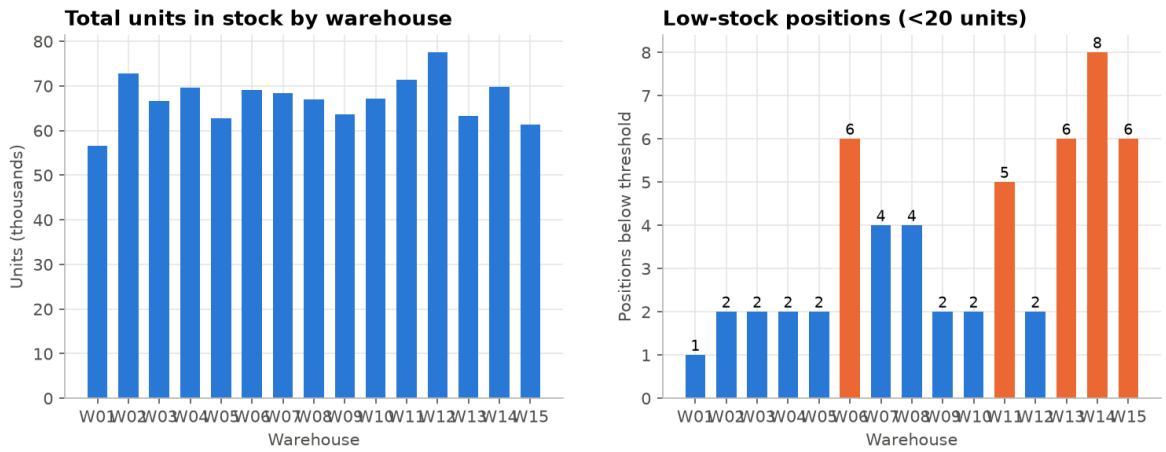


Figure 14. Warehouse vs inventory levels. Stock held ranges from 56.5K (W01) to 77.5K (W12) units. Low-stock positions cluster in W14, W06, W13 (20 of 54).

Business implication: *Low-stock positions are concentrated in a handful of warehouses (a few sites hold over a third of them), so replenishment should be targeted there first rather than raised network-wide.*

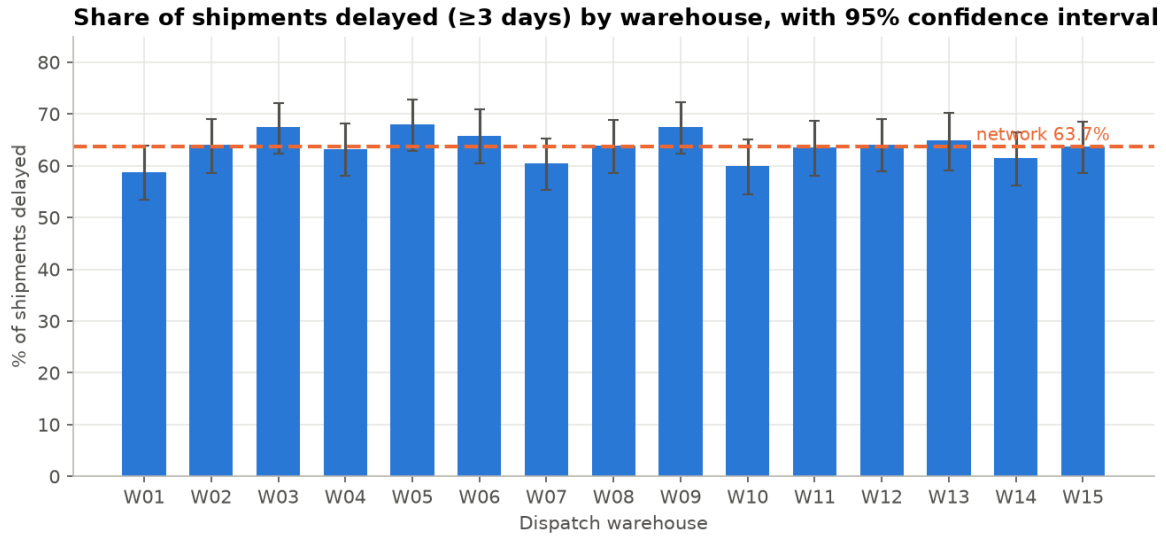


Figure 15. Delivery performance by warehouse. Delay rates run from 58.7% (W01) to 68.0% (W05), but every warehouse's confidence interval overlaps the network average of 63.7% (χ^2 p = 0.33).

Business implication: *Warehouse-to-warehouse differences are statistically indistinguishable from chance. Management should NOT single out a "bad" warehouse; the delay problem is network-wide and points to a shared upstream cause (carrier SLAs, dispatch planning or the promise-date rule).*

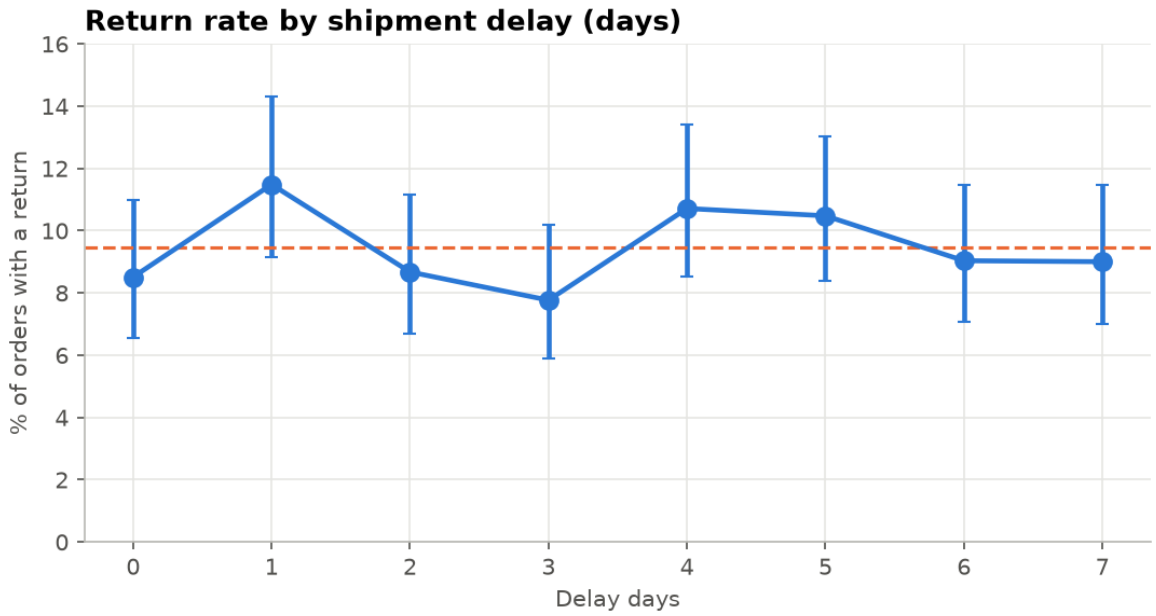


Figure 16. Delay days vs return rate. Return rate does not rise with delay: 8.5% at 0 days vs 9.0% at 7 days; the correlation is ρ = -0.000 (p = 0.98).

***Business implication:** In this data, late delivery does not measurably drive returns, so cutting delays should be justified by service-level and customer-experience goals rather than by an expected fall in returns.*

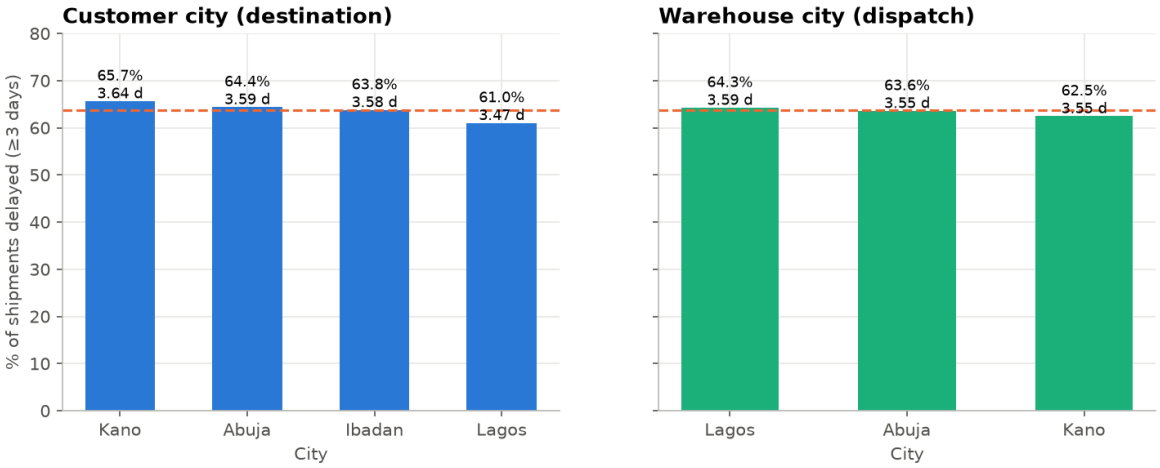


Figure 17. Region vs shipment performance. By destination, Kano has the highest delay rate (65.7%) and Lagos the lowest (61.0%), but the spread is small (χ^2 p = 0.10); by dispatch city the rates are within one point of each other (p = 0.60).

***Business implication:** No region is a significant delay hotspot. Regional delay "league tables" would mislead; the answer to "which region is worst" is "none reliably" — so there is no case for region-specific carrier changes on this evidence.*

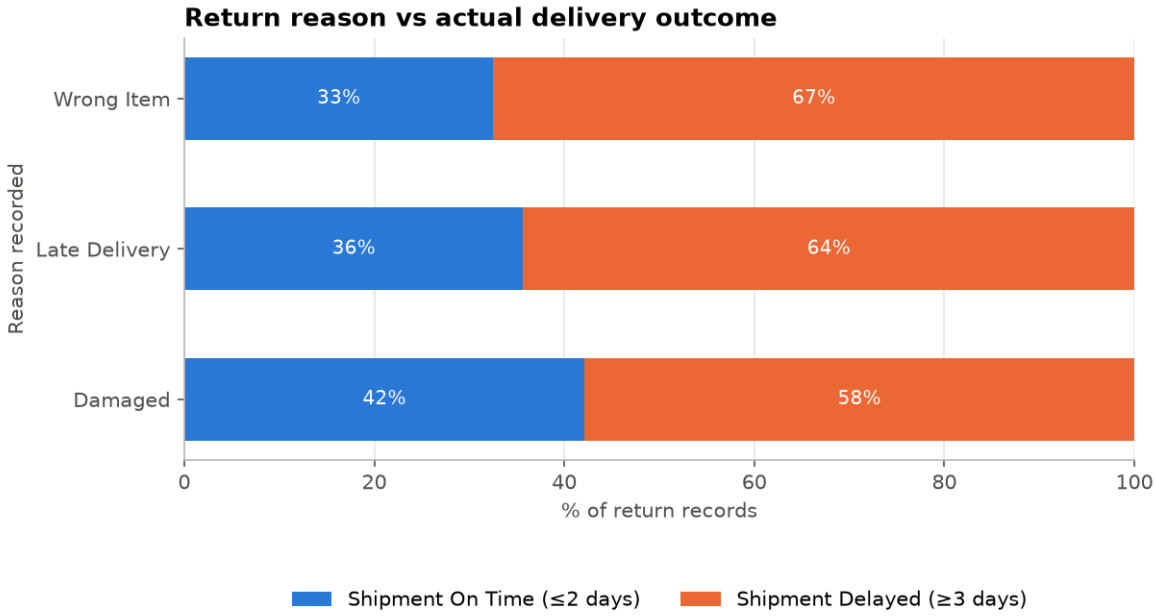


Figure 18. Return reason vs delivery outcome. Only 64% of returns logged as 'Late Delivery' were actually delayed shipments — about the same as 64% of all shipments, and no different from 'Damaged' or 'Wrong Item' returns (χ^2 p = 0.20).

***Business implication:** The 'Late Delivery' reason code is not corroborated by the delivery data: it is either entered inconsistently or the delay field does not reflect what customers experienced. Return reason capture needs an audit and a controlled pick-list.*

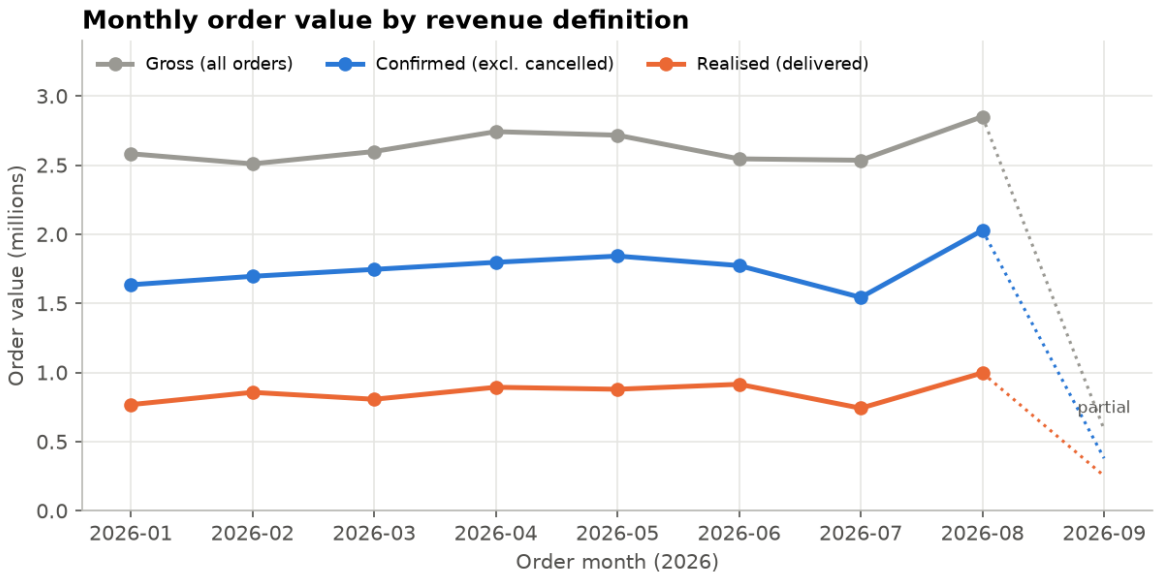


Figure 19. Monthly revenue trend. Confirmed revenue averages 1.76M a month, peaking in 2026-08 (2.03M); there is no sustained growth trend across Jan–Aug. Realised (delivered) value is only 33% of gross.

***Business implication:** Demand is flat, not growing, and the gap between gross and realised value (≈14.56M) is the real prize. September is partial (to 8 Sep) and must not be read as a decline.*

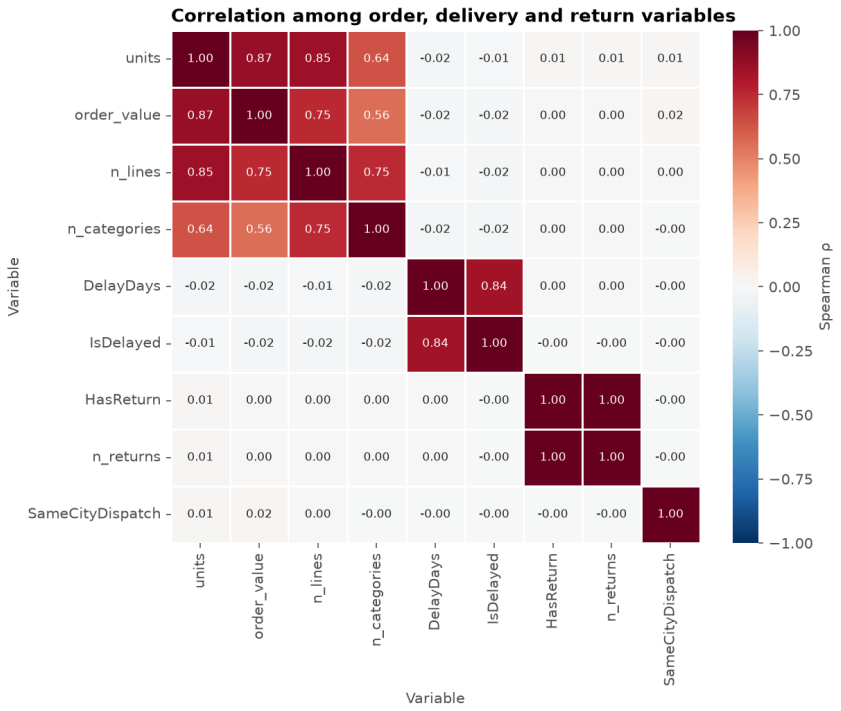


Figure 20. Correlation heatmap. Strong correlations are purely structural (units ↔ order value ↔ lines, delay days ↔ delay flag). Excluding those, the largest relationship between any operational variable pair is |ρ| = 0.02 (order_value vs DelayDays) — effectively zero.

***Business implication:** Delivery delay, returns and basket characteristics are statistically unrelated in this dataset. That caps the predictive power any model can extract from these fields and tells SwiftLog that better predictors (carrier, route, dispatch/promise dates, picker, product-level defect flags) must be captured.*

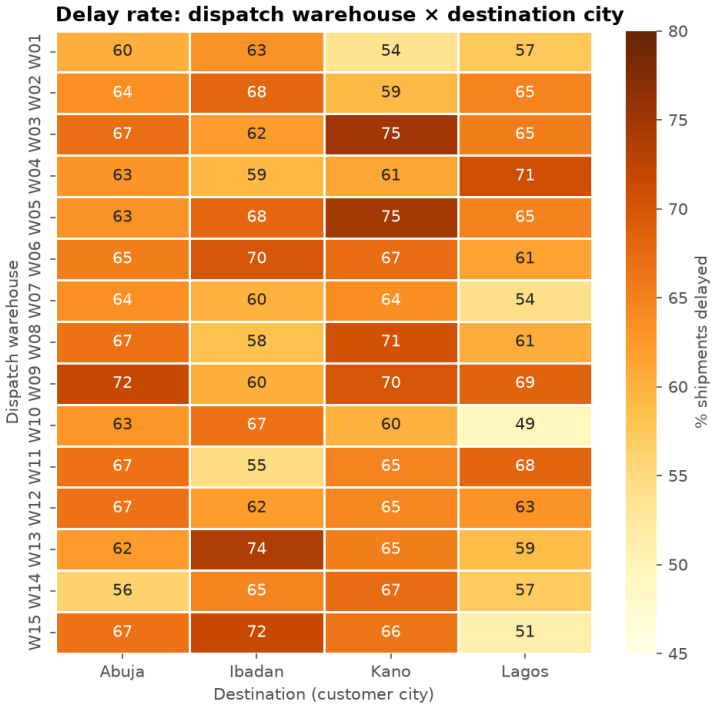


Figure 21. Warehouse × destination delay heatmap. Across the 60 warehouse–city lanes the delay rate ranges from 49% to 75%, but each lane has only ~84 shipments, so most of the spread is sampling noise (expected ±10 pts).

***Business implication:** Lane-level league tables are unreliable at this volume. Before acting on any individual lane, accumulate more shipments or pool lanes (e.g. by region). Monitor with control limits rather than raw ranks.*

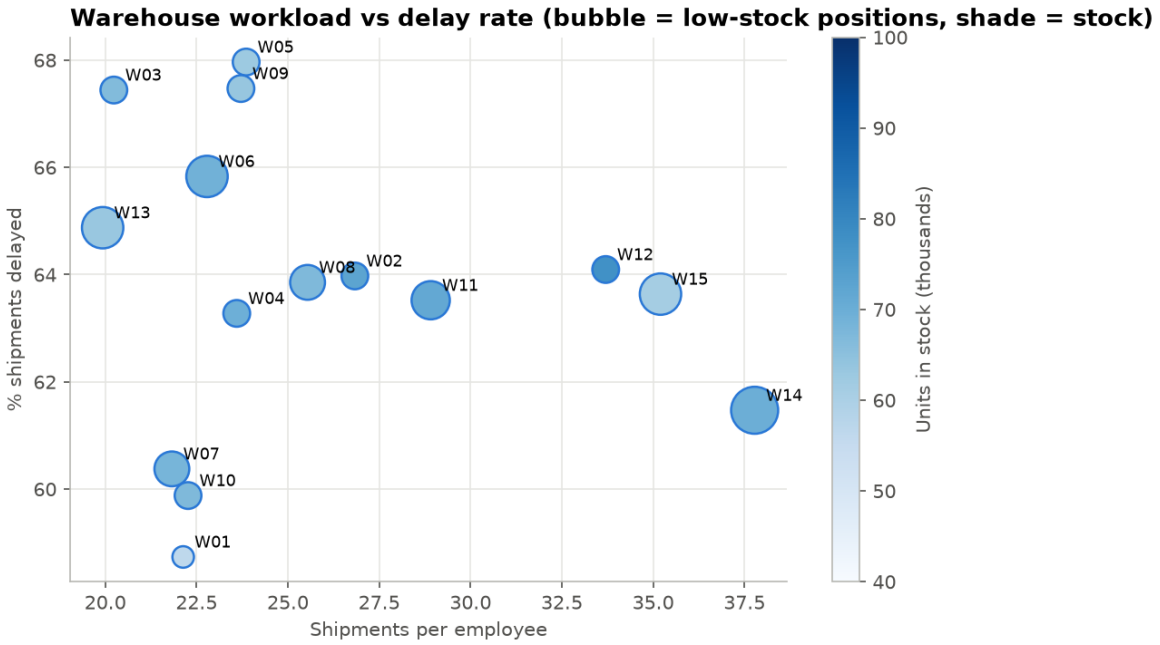


Figure 22. Workload, delay and stock by warehouse. Heavier workload per employee is not associated with more delays (ρ = -0.02, p = 0.93, n = 15) and only weakly with low-stock positions (ρ = 0.27, p = 0.32; not significant).

***Business implication:** The link between workload and low stock is weak and not statistically significant with only 15 warehouses, so it should be monitored rather than acted on. Workload does not explain delays at all, which again points to network-wide causes.*

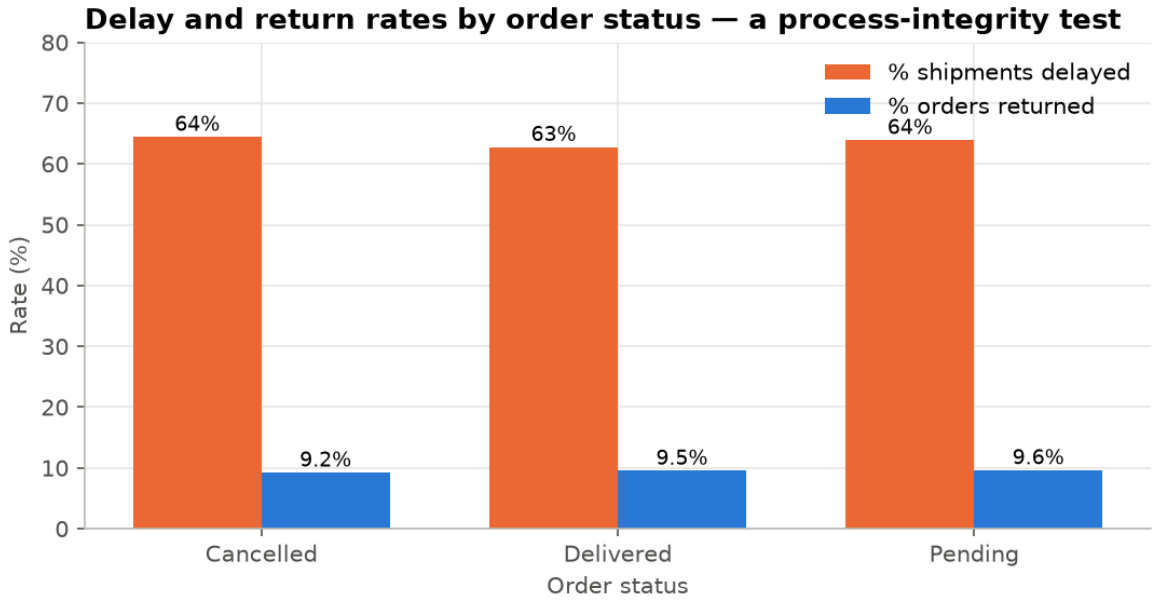


Figure 23. Order status vs delay and returns. Cancelled and Pending orders show the same delay (64% / 64%) and return rates (9.2% / 9.6%) as Delivered orders: 1,626 shipments and 161 returns are recorded against Cancelled orders.

***Business implication:** A cancelled order cannot be shipped late or returned. Either status is not being updated (systems out of sync) or shipment/return records are mis-keyed. Until fixed, every delivery KPI is contaminated — reconciling order status with shipment and return events is a prerequisite for trustworthy analytics and ML.*

5. Statistical Analysis

Order value and units are right-skewed, so non-parametric tests were used. Ten hypothesis tests were run at alpha 0.05 with effect sizes. Bonferroni threshold for ten tests: 0.0050.

Test	Method	p-value	Effect size	Significant?
T1 Delay vs warehouse	Chi-square independence	0.328	Cramér's V = 0.056	No
T2 Delay vs destination city	Chi-square independence	0.099	Cramér's V = 0.035	No
T3 Returns vs delay	Chi-square independence	0.928	Cramér's V = 0.002	No
T4 Order value: delayed vs on-time	Mann–Whitney U	0.160	rank-biserial r = 0.024	No
T5 Order value by dominant category	Kruskal–Wallis	0.100	epsilon² = 0.001	No
T6 Order value by status	Kruskal–Wallis	0.130	epsilon² = 0.001	No
T7 Delay: same-city vs cross-city dispatch	Chi-square independence	0.936	Cramér's V = 0.002	No
T8 Cancellation vs destination city	Chi-square independence	0.963	Cramér's V = 0.012	No
T9 Basket size vs delay days	Spearman correlation	0.229	rho = -0.017	No
T10 Returns vs warehouse	Chi-square independence	0.339	Cramér's V = 0.056	No

Result: 0 of 10 tests were significant at 0.05 and 0 survived correction. Every effect size is negligible (Cramér's V below 0.06). Business reading: delay, returns, order value and warehouse or region show no reliable relationship in this extract. 95% confidence intervals: delayed share 62.4–65.0%, return share 8.7–10.3%.

6. Feature Engineering

The feature table has 5,000 orders and 82 columns (Data/Feature_Engineered_Dataset.csv). Customer and warehouse history use only information available before each order, and post-outcome fields (delay days, returns, status) are never used as predictors, which prevents leakage. Missing values are structural (first orders have no history) and are median-imputed with explicit missing-flag columns.

Group	Feature	Why created	Business value	ML value
Time	IsWeekend, IsMonthEnd, DaysSinceStart, Month_sin/cos	Calendar cycles and trend	Roster and peak planning	Seasonality without leakage; time-split index
Basket	value_per_unit, log_order_value, units_per_line, is_multi_category, is_high_value, is_bulk	Describe order size, price level and pick complexity	Spot premium or complex orders	Scale-free, less skewed predictors
Customer	cust_prior_orders, cust_prior_value, cust_avg_prior_value, days_since_prev_order, is_repeat_customer, value_vs_cust_avg	RFM-style history before the order	Loyalty, churn warning, lifetime value	Behaviour signals with no look-ahead
Warehouse	wh_staff, wh_total_stock, wh_low_stock_share, wh_workload_per_emp, wh_prior_delay_rate, wh_orders_last7d, wh_load_index	Capacity proxies and rolling performance	Staffing and replenishment monitoring	Static and dynamic site-load signals
Route	SameCityDispatch, route	Local vs cross-city fulfilment	Lane reporting	Distance proxy (no coordinates available)
Encoding	one-hot city and category columns, Status_code	Models cannot use raw text	None directly	One-hot avoids false ordering; few levels

Screening: only 2 of 24 candidate features reached $p < 0.05$ against a target (about 2 expected by chance), and the strongest rank correlation was 0.034. Baseline models (logistic regression, random forest, gradient boosting; train on first 70% of dates, test on last 30%) gave test AUC of 0.52 for delay and 0.52 for returns, i.e. no predictive power. The honest conclusion is that the engineered features are sound and ready, but the source data lacks the drivers of delay.

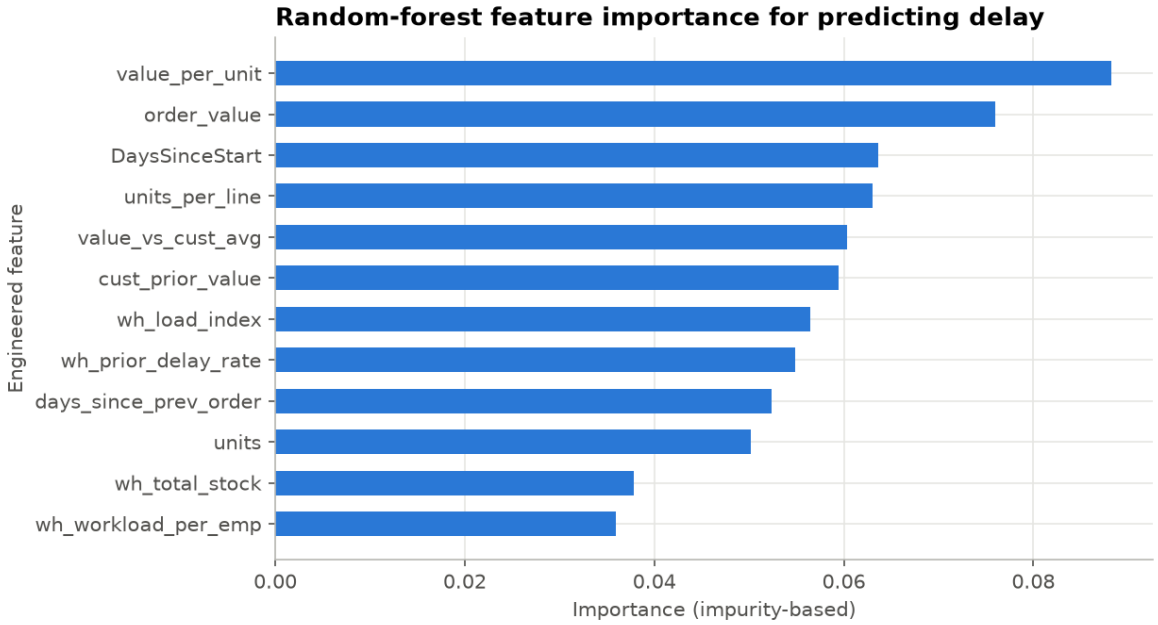


Figure 24. Feature importance (delay model). Importance is spread thinly across features, led by value_per_unit and order_value; with a test AUC of only 0.52 the ranking mostly reflects noise rather than real drivers.

***Business implication:** Today's extract cannot predict delays. Useful ML needs new inputs: carrier, route distance, promised vs actual dates, picker/shift, weather and stock availability at pick time.*

7. Segmentation Results

Customers were segmented by log-scaled recency, frequency and monetary value using K-Means with k = 4 (silhouette 0.30, modest, so segments are planning buckets rather than sharp groups). 14 customers whose confirmed orders carry no line-item value were excluded and reported separately. Products were classed A/B/C by revenue, and warehouses were clustered on workload, delay, stock risk and returns (silhouette 0.24).

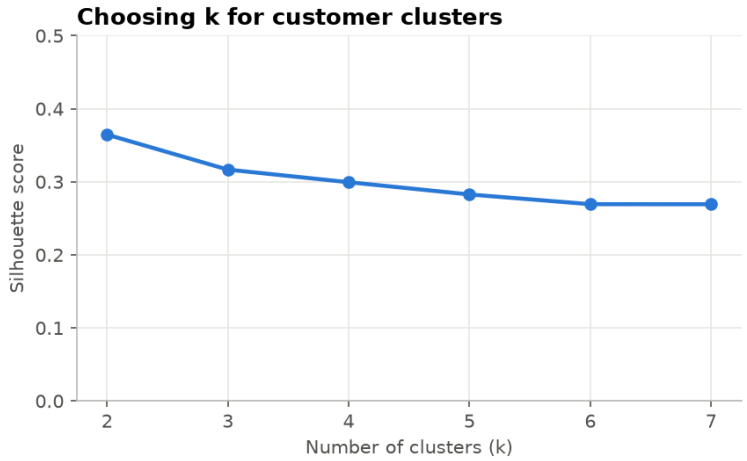


Figure 25. Cluster count selection. Silhouette scores are modest (best 0.36 at k = 2); anything below ~0.5 signals clusters that overlap rather than form distinct groups.

Business implication: *Customer segments here are useful as planning buckets (high/low value) but are not natural, sharply separated groups — avoid over-interpreting them.*

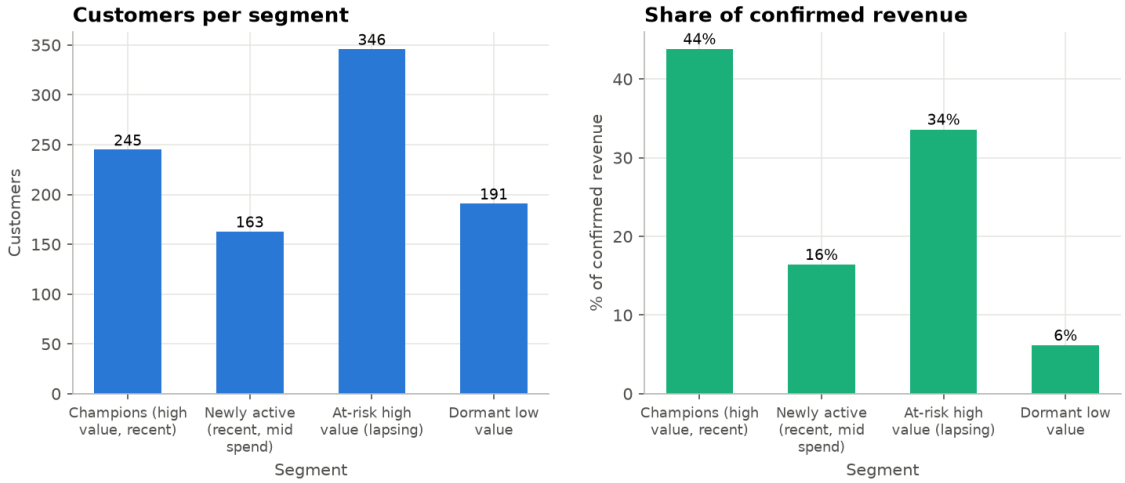


Figure 26. Customer segments (RFM + K-Means). Champions (245 customers) average 25.8K each and hold 44% of confirmed revenue. Segments differ in recency and frequency but all share the same delay rate (≈65%).

Business implication: *Retention effort should go to the high-value customers whose last order is old (lapsing) before they are lost; every segment suffers the same delivery experience, so service reliability is a retention risk for all of them.*

Segment	Customers	Avg recency (days)	Orders	Avg value	% revenue
Champions (high value, recent)	245	39	5.7	25,827	44%
At-risk high value (lapsing)	346	86	3.1	14,018	34%
Newly active (recent, mid spend)	163	9	3.7	14,539	16%
Dormant low value	191	94	1.5	4,653	6%

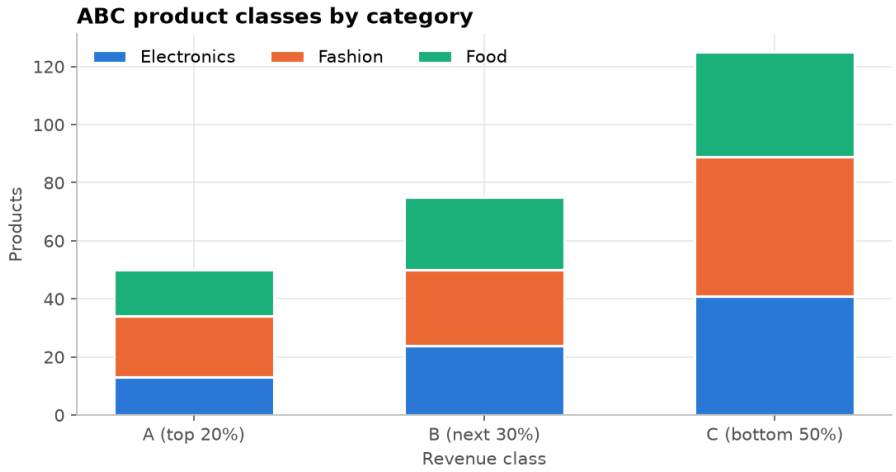


Figure 27. Product ABC segmentation. Class A (top 20%, 50 products) earns 36.0% of confirmed revenue; class C (bottom 50%) still earns 27.3%. All three categories appear in every class.

Business implication: *ABC differences are mild, so a tail clean-up would give up real revenue. Manage class A for availability (never stock out), and review class C on margin and return data before deleting any line.*

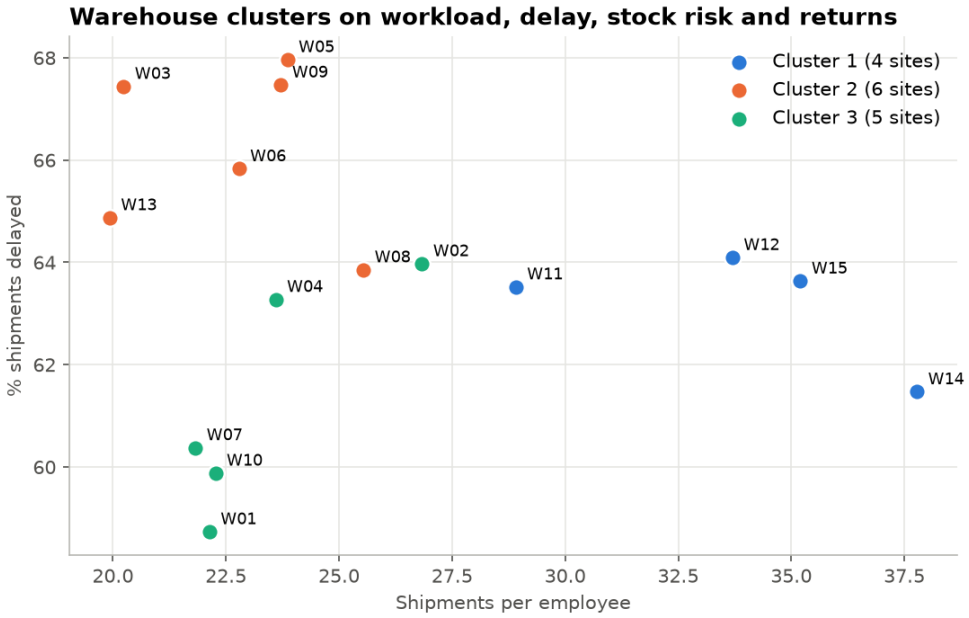


Figure 28. Warehouse performance clusters. Three warehouse clusters emerge (silhouette 0.24). Best composite reliability: W01; weakest: W15. Because delay differences are statistically insignificant, clusters are driven mostly by workload and stock risk.

Business implication: *Use the composite Reliability Score as a monitoring dashboard, not a ranking for blame. It is the best supplier-performance substitute available until supplier IDs are captured on inventory and shipments.*

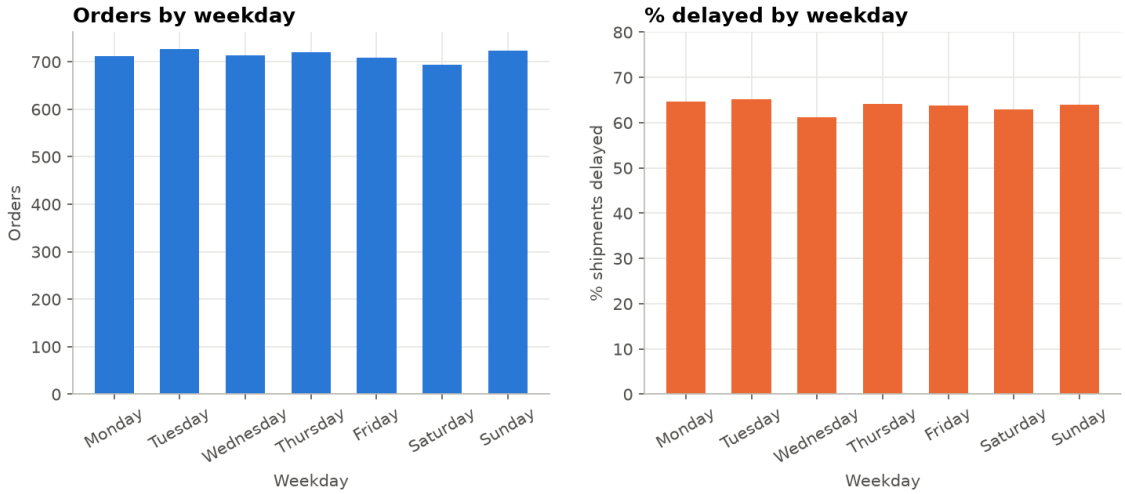


Figure 29. Weekday patterns. Order volume is flat across weekdays (694–726 orders) and delay rates stay within 61%–65%; no weekday effect.

Business implication: *There is no peak day to staff for, and no day where delays are worse; weekly roster changes would not help. Orders also arrive every day of the week including weekends, which is unusual and worth confirming with the order system owners.*

Supplier segmentation is not possible because no supplier key exists on any operational table. The warehouse Reliability Score (z-score composite of delay, return rate and low-stock positions) is the best available substitute.

8. Business Insights

1. Which customer segments generate the highest business value?

Champions (high value, recent) — 245 customers averaging 25.8K each, 44% of confirmed revenue. A second value pool is the lapsing high-value group (346 customers, 34% of revenue, last order ≈86 days ago). Revenue is not concentrated in a few accounts (top 20% of customers = 41%, Gini 0.38). Evidence: charts 10, 11, 26.

2. Which products contribute most to revenue?

Fashion is the top category (38% of confirmed revenue, partly because it has the most SKUs); the top products are P085, P040, P025. But concentration is low: top 20% of products = 36% of revenue and 145 of 250 products are needed for 80%. Rankings also shift with the status filter (only 6/10 top products are the same on delivered-only revenue). Evidence: charts 12, 13, 27.

3. Which suppliers consistently underperform?

This cannot be answered. The Suppliers table lists 150 IDs and names only; no SupplierID exists on Products, Inventory, Shipments or Orders, so no delivery, quality or return outcome can be traced to a supplier. We did not fabricate a link. Substitute: warehouse reliability score (chart 28). Fix: add SupplierID to Products/Inventory receipts and capture PO promised vs received dates.

4. Which warehouses require operational improvements?

Delay differences between warehouses are not statistically significant (χ^2 p = 0.33), so no warehouse can be blamed for delays. The operational gaps are staffing and stock: highest shipments per employee are W14, W15, W12 (up to 38 vs 20), and low-stock positions cluster in W14, W06, W13. Composite reliability: best W01, weakest W15. Evidence: charts 8, 14, 15, 22, 28.

5. Which products have the highest return rates?

Highest observed: P032, P017, P108 (up to 21.3% vs 9.5% overall), but with ~59 lines per product this is mostly noise: 0 products exceed ±3 SE (≈0.7 expected by chance) and category does not matter (p = 0.96). Reason mix: Late Delivery 185, Wrong Item 163, Damaged 152. No product should be delisted on return rate alone. Evidence: chart 7.

6. Which regions experience the greatest delivery delays?

Kano has the highest delay rate (65.7%) and Lagos the lowest (61.0%), but the difference is not significant (p = 0.10). Honest answer: no region is reliably worse; the network average is 63.7%. Evidence: charts 17, 21.

7. What operational bottlenecks have you identified?

(1) Network-wide lateness: 63.7% of shipments are ≥3 days late and delay is uniform 0–7 days. (2) Order conversion: only 33.2% of orders are Delivered; 32.5% Cancelled (7.23M) and 34.3% still Pending. (3) Process integrity: 1,626 shipments are recorded against Cancelled orders. (4) Staffing imbalance (1.9× workload spread). (5) Order-capture gaps: 236 orders without line items. Evidence: charts 3, 6, 8, 23.

8. Which variables are most suitable for predictive modelling?

Candidate targets: IsDelayed, DelayDays, HasReturn. Engineered features (customer history, warehouse rolling delay, load, basket) were tested, but the best time-split AUC was 0.52 for delay and 0.52 for returns — essentially chance; 2 of 24 features pass p<0.05 (≈2 by chance). So current variables are NOT suitable for delay/return prediction. Revenue, order volume and customer recency/frequency are the modelling-ready quantities (forecasting, churn). Evidence: chart 24.

9. What hidden patterns did your analysis uncover?

(a) Statuses do not drive outcomes: Cancelled and Pending orders have the same delay/return rates as Delivered ones, implying the status field and shipment/return events are out of sync. (b) 'Late Delivery' returns are no more likely to be actually delayed (64%) than any other return. (c) Delay is unrelated to value, basket, route, weekday or returns (max |ρ| 0.02). (d) Revenue is flat across Jan–Aug with no growth trend. (e) Ibadan has the largest customer base but no warehouse. Evidence: charts 18, 20, 23.

10. What recommendations would you make to improve SwiftLog's operations?

15 prioritised recommendations — see table below; top priorities: status reconciliation, network-wide on-time programme, cancellation funnel, data capture, win-back.

9. Recommendations

Fifteen strategic recommendations, each tied to evidence in this report.

#	Recommendation	Evidence	Priority	Expected impact
1	Reconcile order status with shipment and return events	1,626 shipments and 161 returns sit on Cancelled orders; delay KPIs are contaminated	High	Trusted KPIs; prerequisite for any ML
2	Launch a network-wide on-time delivery programme (carrier SLAs, dispatch cut-offs, promise-date review)	63.7% late, uniform across sites/regions — a systemic cause	High	Each 10-point cut in lateness lifts on-time rate to ~46%
3	Attack the cancellation/pending leak with a funnel review (stock confirmation, payment, address checks)	32.5% cancelled, 7.23M order value; only 33.2% delivered	High	Recovering 10% of cancelled value ≈ 722.9K
4	Rebalance warehouse staffing toward overloaded sites (W14, W15, W12)	1.9× spread in shipments per employee with near-equal volumes	Medium	Better utilisation without extra headcount
5	Targeted replenishment at low-stock hotspots (W14, W06, W13)	54 positions below 20 units, concentrated in few sites	Medium	Fewer stock-outs without raising total stock
6	Win-back programme for lapsing high-value customers	346 customers, 34% of revenue, ≈86 days since last order	High	Protects about a third of confirmed revenue
7	Introduce a loyalty/frequency scheme across the whole base rather than key-account management	Top 20% of customers = 41% of revenue (Gini 0.38); no whales	Medium	Raises orders per customer (now 5.0)
8	Assess an Ibadan fulfilment point	Largest customer city (26%) with no warehouse; all Ibadan orders ship cross-city	Medium	Possible shorter lead times; needs cost-benefit (distance data not available)
9	Capture the missing data: SupplierID, carrier, promised/dispatch/delivery dates, warehouse capacity, return product	Supplier performance, capacity utilisation and ML are impossible today	High	Unlocks supplier scorecards and delay prediction
10	Fix order capture: block orders without line items and fix the empty-basket cohort	236 orders (4.7%) have no items; 14 customers with no valued orders	Medium	Complete revenue visibility
11	Replace free-text return reasons with a controlled pick-list and audit "Late Delivery" coding	Only 64% of 'Late Delivery' returns were delayed shipments	Medium	Reliable root-cause for returns
12	Tackle controllable returns (wrong item, damaged) through pick-accuracy checks and packaging	Wrong Item + Damaged = 63% of returns; 9.5% of orders affected	Medium	Lower return handling cost
13	Manage the range by revenue per SKU and margin, not total revenue; do not cut the tail yet	145 of 250 products needed for 80% of revenue; class C still earns 27%	Low	Avoids losing tail revenue
14	Monitor delays with control charts at region level, not warehouse/lane league tables	Lane samples ≈84 shipments; most spread is noise	Low	Avoids acting on noise
15	Build future ML (delay-risk scoring, demand forecast, churn) only after new data fields exist	Current best AUC 0.52	Medium	Avoids deploying a useless model

10. Conclusion

SwiftLog has a stable but flat order book and a serious, network-wide delivery problem: 64% of shipments are late, only 33% of orders reach Delivered status, and the records themselves are internally inconsistent. The analysis found no warehouse, region, product or customer group that explains the delays, which means the cause lies in factors that are not recorded today.

The most valuable next step is therefore data, not modelling: capture supplier, carrier, promised and actual dates, capacity and controlled return reasons, and reconcile order status. With those in place, delay-risk scoring, demand forecasting and churn prediction become feasible. The feature pipeline and cleaning functions built here can be re-run unchanged on richer data.

Limitations

- Order dates cover 1 Jan – 8 Sep 2026; September is partial.
- No currency, cost, margin, supplier link, carrier, capacity or ship/delivery dates in the data; revenue is order value from unit price × quantity.
- Order status is inconsistent with shipments/returns, so "realised" revenue is a conservative lower bound.
- The synthetic-looking uniformity of delay, returns and values limits predictive and segmentation power; tests are many, so only Bonferroni-surviving findings should drive investment (none did).
- Warehouse capacity is unknown, so workload per employee is a proxy not utilisation.

Future ML opportunities

- Delivery-delay risk scoring once carrier, route distance and promised dates are captured
- Demand and revenue forecasting by category and city (needs a longer history than 8 months)
- Customer churn / win-back propensity from the RFM features built here
- Return-risk scoring once product-level defect and picker data exist
- Anomaly detection on order status vs shipment events to catch process errors early